MIL-S-51210A(EA)
15 September 1976
SUPERSEDING
MIL-S-51210(MU)
14 October 1964

MILITARY SPECIFICATION

SOIL SAMPLING KIT

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a small canvas field kit containing soil sampling equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

DL5-77-458 - Soil Sampling Kit.

FSC 6665

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

PHARMACOPEIA CONVENTION PUBLICATION

The Pharmacopeia of the United States, XIV Edition.

(Application for copies should be addressed to the Mack Publishing Co., Easton, Pennsylvania 10842.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-77-458 or on subsidiary drawings, shall conform to the specifications thereon, or the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the kit shall conform to the specifications and drawings listed on Drawing DL5-77-458 and the subsidiary drawings.

3.2 Manufacture and assembly. The kit shall be manufactured and assembled as specified on Drawing D5-77-458.

3.2.1 Stitching. All rows of stitching shall be straight or regular lines, free from loose stitches or thread loops. When stitching is not of continuous thread, each thread or break shall be backstitched, wherever practicable, for no less than 2 inches. All seams and hems shall be backstitched at the ends with two or three stitches to prevent raveling.

3.2.2 Clip. When attaching clips, the duck shall be inserted the full length of the clip. The pressure used in assembling the clip to the duck shall be such as not to cut, break, or weaken the duck. The clip shall form a smooth, flat, tight and compact assembly.

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3.3 Bag durability. The soil extraction bags and collection bags shall not puncture, tear or burst, and the tie strings shall not break when tested as specified in 4.4.4.1.

3.4 Cotton-tipped swab sterility. The cotton-tipped swabs shall be sterile and shall exhibit no growth of organisms when tested as specified in 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of soil sampling kits shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. Workmanship shall be of such nature that the kit, carrier and components are free from damage (rips, tears, or punctures), defects or foreign matter (grease, oil or viscous material).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five soil sampling kits shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedures.

4.3.2.1 For examination. Each preproduction kit shall be examined in accordance with 4.4.3.1.

4.3.2.1 For tests. Five collection bags and two extraction bags shall be taken from the preproduction sample and tested in accordance with 4.4.4.1. Six cotton-tipped swabs shall be taken from the preproduction sample and divided into two equal groups, Group A and Group B, and tested in accordance with 4.4.4.2.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the soil kits produced by one manufacturer, under essentially the same manufacturing conditions and with no change in materials. However, no more than one lot of each component shall be represented in one lot of finished kits.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Six cotton-tipped swabs, five collection bags, and two extraction bags shall be taken at random from each lot. The six cotton-tipped swabs for sterility testing shall be separated into two groups, group A containing three swabs and group B containing three swabs.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample kits shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For test. Sample bags and cotton-tipped swabs shall be tested as specified in 4.4.4. The acceptance number is zero.

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4.4.3.3 Classification of defects.

(a) Soil sampling kit (Dwg D5-77-458).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrectly assembled
102	Bottle contents missing or incorrect
103	Hardware nonfunctioning
104	Stitches loose or looped
105	Broken stitches not backstitched
106	Marking missing, incorrect, or illegible
107	Workmanship (3.6)

(b) Bag, soil extraction (Dwg C5-77-435)

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Center heat seal missing

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Bag durability.

(a) Collection bag. Fill the bag with 250 grams of dry sand and roll the end three times and tie off. Drop the bag 20 times from a height of 6 feet onto a concrete surface so that it lands on the large flat side but not on the edge. Examine the bag for compliance with 3.3.

(b) Extraction bag. Fold and tie with string the small opening of the extraction bag. Test as indicated in 4.4.4.1(a).

4.4.4.2 Cotton-tipped swab sterility. Conduct these tests under aseptic conditions as follows:

(a) Group A. Using 20 by 150 millimeters (mm) test tubes fill 20 milliliters of fluid thioglycollate medium (prepared as specified in the Pharmacopeia of the United States, XIV Edition, pages 758 - 759, and

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complying with all requirements for pH, sterility, and purity as specified in that publication) into each of four test tubes and sterilize in an autoclave at 121.5° C (15 PSIG) for 20 minutes. Cool at once to 25° C. Transfer each of three cotton-tipped swabs from the envelopes, into three test tubes of sterile medium, placing one swab in each tube. Omit swab from the fourth test tube of sterile medium and use as a control. Incubate the tubes at a temperature of 37.0 ± 0.5° C for a period of 7 days. Visually examine the tubes periodically and on the 7th day for growth or cloudiness which is evidence of growth. Confirm any growth by microscopic examination of stained slides.

(b) Group B. Using 20 by 150 mm test tubes fill 20 ml of Sabouraud Liquid medium (modified) (prepared as specified in the Pharmacopeia of the United States, XIB Edition, page 760, and complying with all requirements for pH, sterility, and purity as specified in the referenced publication) into each of four test tubes and sterilize in an autoclave at 121.5° C (15 psig) for 20 minutes. Cool at once to 25° C. Transfer each of three cotton-tipped swabs from the envelopes into three test tubes of sterile medium, placing one swab in each tube. Use the fourth test tube of sterile medium without a swab as a control. Incubate the tubes at a temperature of 23.5 ± 1.5° C for a period of 15 days. Visually examine the tubes for growth or cloudiness which is evidence of growth periodically for 7 days with a final examination on the 15th day. Confirm any growth by microscopic examination of stained slides.

4.4.5 Acceptance/rejection criteria. Failure of any sample to comply with the requirement specified in 3.4 shall be cause for rejection of the lot represented. Rejected lots may again be sterilized and submitted for acceptance.

5. PREPARATION FOR DELIVERY

5.1 Packaging, commercial. The kit shall be packaged in accordance with FED-STD-356.

5.2 Packing, commercial. The kit, packaged as specified in 5.1, shall be packed in accordance with FED-STD-356.

5.3 Marking. In addition to any marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. The kit covered by this specification is a component of the M19 CBR Agent Sampling and Analyzing Kit and is intended to be used when soil samples contaminated with CBR agents are to be collected for transportation to a laboratory for examination.

6.2 Ordering data. Procurement document should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6665-A378

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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